

Element E

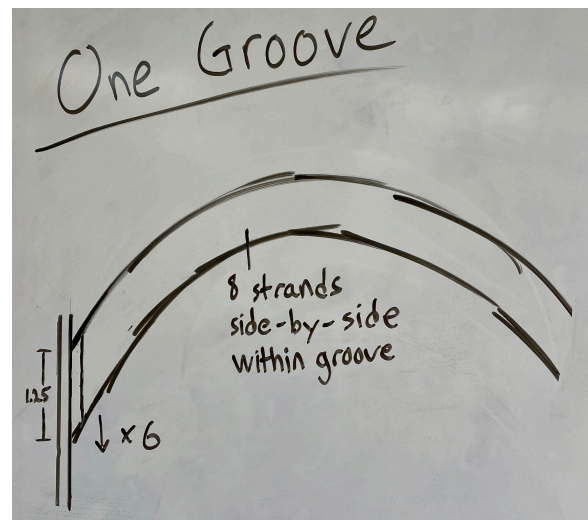
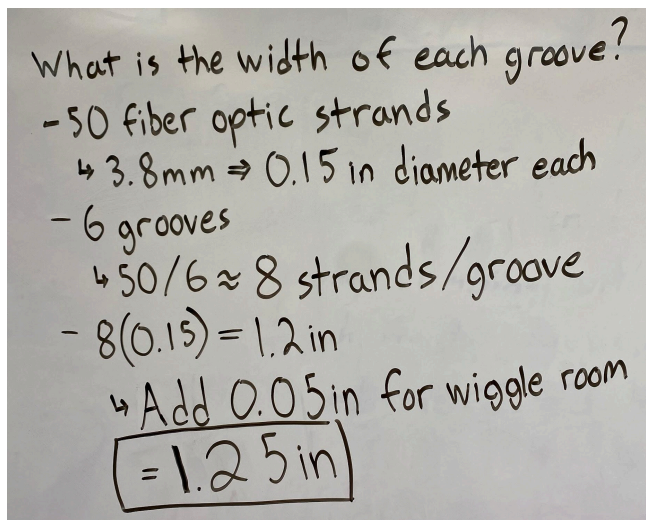
STEM Principles:

Science - Using colors to affect the mood of the room is related to chemistry and biology. The hormones that change one's mood are caused by chemical changes in the brain.

Technology - The Fiber Optic light machine, the cornerstone of our project, is a piece of technology.

Engineering - In designing the wave board, we employed the steps of the engineering design process. We identified the lack of visual sensory as the problem, created a list of constraints, then we created a morphological chart. Following this, we created conceptual designs, a decision matrix, and then choose the best one based on our constraints.

Mathematics - We had to derive what width each groove of the board should be, so we used mathematic conversions and equations:



Evidence of Design Being Reviewed by Experts:

Expert	Feedback	What We Changed
Nate Johnson (Associate vehicle engineer for exponent with a M.S. in mechanical & aerospace engineering)	Mr. Johnson liked our project and the wave design, but he wanted us to further our research into fiber optics to make sure we could use them accordingly. Additionally, he encouraged us to increase our communication with Ms. Brennan in order to ensure that we're on the same page.	Before discussing our designs with Mr. Johnson, we hadn't established the exact dimensions or location of the waveboard in the sensory room with Ms. Brennan, or whether she would have to make any design changes around the rooms. We also had not calculated the length or number of fiber optic wave

		strands we would need for our design blueprint. After our discussion, we reached out to Ms. Brennan, and coordinated a time to visit the sensory room at Whitemarsh, where we shared our design idea and established a design that fit her parameters and vision for the project. We also calculated the diameter and further the number of fiber optic strands we would need for our design.
Dr. Joseph Sarver (Professor at Drexel University with a Ph.D. in biomedical engineering and a B.S. in mechanical engineering)	“Fiber optic wave board reasoning seems sound, would suggest your design should allow a teacher to manipulate performance (which colors, how fast they change, do they dim/brighten, etc), which will allow them to adjust and see what works best”	Found fiber optics that came with a preprogrammed remote. This allows any teacher/aid to easily change the color, speed, and brightness of the fiber optics.
Dr. Jamie Dougherty (Associate professor at Drexel University with a Ph.D. in biomedical engineering)	“I think the fiber optic wave board sounds like a clever idea. (I wonder if they added a component that sticks out of the board, something like this they may be able to have something students can really touch, and you could probably get something off-the shelf that you modify with a button of some sort. Just a thought.) I do think that the slower changing wave component is really smart, if you wanted more interactivity maybe you could code it to change faster or slower? I don't know much about that, I honestly think the slow changing wave of colors sounds fantastic. As a person with a kid who is easily stimulated, I think that would be great. Working with the stakeholders can be challenging, but I suspect	Kept our design simple so Mrs. Bennen would have no trouble understanding what we are trying to build. It also makes it easier for her to make suggestions on our design so it can best fit her sensory room. Found fiber optics that allowed the speed of the lights to be changed easily using a remote.

	they may not know exactly what they want, so I think keeping it simple is the best scenario. Keep in mind that what they have now is very limited, so anything you make will be an improvement!"	
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